

Textbook of Medical Physiology: Guyton Principles

Physiology — MBBS 1st Year

Standard Core Curriculum for Medical Undergraduates • Comprehensive Clinical Reference

Curriculum Year	MBBS 1st Year
Medical Subject	Physiology
Target Audience	MBBS Undergraduates, Postgraduates, USMLE/PLAB Aspirants
Format	Fully Indexed Textbook with Chapter, Topic, and Page Referencing
Accreditation	Standard Medical Council Curriculum Guidelines

Chapter 1: Cardiac Electrophysiology, Hemodynamics and Blood-Brain Barrier

Phases of the Cardiac Cycle, Pressure-Volume Loops and Wiggers Diagram

The cardiac cycle consists of mechanical and electrical events from one heartbeat to the next. At resting heart rate 75 bpm, each cycle lasts 0.8s (systole 0.3s; diastole 0.5s). Phase 1 is Isovolumetric Contraction: rising ventricular pressure abruptly closes AV valves (mitral/tricuspid), producing the First Heart Sound (S1). All valves remain closed; ventricular volume stays at End-Diastolic Volume (EDV, ~120 mL).

Phase 2 is Rapid Ventricular Ejection: LV pressure exceeds aortic diastolic pressure (~80 mmHg), opening the aortic valve and rapidly ejecting two-thirds of stroke volume. Phase 3 is Reduced Ejection. Phase 4 is Isovolumetric Relaxation: falling ventricular pressure below aortic pressure leads to abrupt aortic valve closure, generating the Second Heart Sound (S2) and dicrotic notch at End-Systolic Volume (ESV, ~50 mL).

Phase 5 is Rapid Ventricular Filling: AV valves open as ventricular pressure falls below atrial pressure. Rapid passive inflow accounts for 70-80% of ventricular filling. Rush of blood into a stiff or volume-overloaded ventricle produces a Third Heart Sound (S3), hallmark of heart failure.

Clinical Pearl: S1 corresponds to AV valve closure at onset of isovolumetric contraction; S2 corresponds to semilunar valve closure at onset of isovolumetric relaxation. The Wiggers diagram synchronizes ECG, pressures, volumes, and heart sounds.

Aortic / Pulmonic Valve	Ventricular Volume
closed	Constant at EDV (~120 mL)
open	Rapidly Decreasing
closed	Constant at ESV (~50 mL)
closed	Rapidly Increasing

Chapter 1: Cardiac Electrophysiology, Hemodynamics and Blood-Brain Barrier

Determinants of Cardiac Output, Frank-Starling Law and Hemodynamics

Cardiac Output (CO) is the volume of blood pumped per minute: $CO = \text{Stroke Volume (SV)} \times \text{Heart Rate (HR)}$, averaging 5.0 L/min in adults. Stroke volume is governed by three factors: Preload, Afterload, and Inotropism (contractility).

The Frank-Starling Law states that within physiological limits, ventricular contraction force is proportional to initial resting length of muscle fibers. Increased venous return increases EDV (preload), stretching myocardial sarcomeres toward optimal actin-myosin overlap (~2.2 micrometers). This enhances troponin C affinity for calcium, generating greater stroke volume.

Afterload represents the tension or resistance against which the ventricle must contract to eject blood (approximated by Mean Arterial Pressure and Systemic Vascular Resistance). Elevated afterload in hypertension or aortic stenosis decreases stroke volume and increases myocardial oxygen consumption.

Clinical Pearl: The Frank-Starling mechanism matches the output of the right and left ventricles on a beat-to-beat basis, maintaining systemic and pulmonary circulation balance.

	Effect on Stroke Volume	Clinical Modu
	Increased preload increases SV (Frank-Starling)	IV fluids, venou
	Increased afterload decreases SV	Systemic vascu
	Increased inotropy increases SV	Digoxin, dobuta

Chapter 1: Cardiac Electrophysiology, Hemodynamics and Blood-Brain Barrier

Cerebrospinal Fluid (CSF) Secretion, Blood-Brain Barrier and Intracranial Dynamics

Cerebrospinal Fluid (CSF) is an ultrafiltrate of plasma produced primarily by the Choroid Plexuses in the lateral, third, and fourth ventricles at ~0.35 mL/min (~500 mL/day). Total normal CSF volume is approximately 150 mL (renewed 3-4 times daily). CSF flows from lateral ventricles through the foramina of Monro into the third ventricle, cerebral aqueduct of Sylvius, fourth ventricle, and exits via foramina of Luschka and Magendie into the subarachnoid space.

CSF is reabsorbed into the venous circulation via Arachnoid Granulations projecting into the Superior Sagittal Sinus. Normal CSF opening pressure in the lateral decubitus position is 80 to 200 mm H2O. The Blood-Brain Barrier (BBB) is formed by non-fenestrated capillary endothelial cells joined by tight junctions (zonula occludens), supported by pericytes and astrocyte end-feet (astrocytic glia limitans).

In acute bacterial meningitis, bacterial endotoxins and inflammatory cytokines (TNF-alpha, IL-1beta) destroy endothelial tight junctions, causing profound BBB breakdown. Protein and fluid pour into the cerebral parenchyma, generating vasogenic and cytotoxic cerebral edema that dramatically spikes intracranial pressure according to the Monro-Kellie doctrine.

Clinical Pearl: The Blood-Brain Barrier consists of endothelial tight junctions, pericytes, and astrocytic end-feet. Disruption in bacterial meningitis causes vasogenic cerebral edema and marked elevation of opening pressure (>250 mm H2O).

Normal CSF Value	Pathological Shift in Bacterial Meningitis
50 mL (turnover ~3-4x daily)	Normal volume but obstructed reabsorption elevates ICP
0 - 200 mm H2O	Markedly elevated (> 250 - 400 mm H2O)
Strictly impermeable to large molecules/proteins	Severe disruption; massive leak of serum albumin and cytokines
0.35 mL/min (~500 mL/day)	Sustained secretion despite obstructed arachnoid villi

Chapter 2: Respiratory and Renal Physiology

Oxygen-Hemoglobin Dissociation Curve, Bohr and Haldane Effects

Hemoglobin reversibly binds four oxygen molecules, yielding a sigmoidal Oxygen-Hemoglobin Dissociation Curve. P50 is the PaO₂ at 50% saturation, normally ~26.8 mmHg under standard physiological conditions (pH 7.40, temp 37°C, PaCO₂ 40 mmHg).

A Rightward Shift indicates decreased hemoglobin affinity for oxygen, facilitating oxygen unloading at tissues. Causes include acidosis (decreased pH), elevated PaCO₂ (Bohr Effect), hyperthermia, and increased 2,3-bisphosphoglycerate (2,3-BPG).

A Leftward Shift reflects increased affinity of hemoglobin for oxygen, impeding tissue release. Causes include alkalosis, hypothermia, decreased 2,3-BPG, fetal hemoglobin (HbF), and carboxyhemoglobin (carbon monoxide poisoning locking Hb in the high-affinity R state).

Clinical Pearl: Remember: CADET face right! CO₂, Acidosis, 2,3-DPG, Exercise, Temperature all shift the curve to the RIGHT, facilitating oxygen unloading to hypoxic tissues.

	P50 Value	Precipitating
	Increased (>27 mmHg)	Increased H ⁺ (a
	Decreased (<26 mmHg)	Decreased H ⁺ (a

Chapter 2: Respiratory and Renal Physiology

Glomerular Filtration Rate (GFR) Dynamics and Countercurrent System

Glomerular Filtration Rate (GFR) averages 125 mL/min (180 L/day). It is governed by Starling forces: $GFR = K_f * [(P_{gc} - P_{bs}) - (\pi_{gc} - \pi_{bs})]$. Hydrostatic pressure in glomerular capillaries (P_{gc} ~55 mmHg) is the primary driver promoting filtration.

Renal clearance measures plasma volume cleared of a substance per minute: $C_x = (U_x * V) / P_x$. Inulin clearance is the gold-standard measure because it is freely filtered without tubular reabsorption or secretion. Creatinine clearance serves as the clinical surrogate.

The countercurrent multiplier system in the loop of Henle establishes a hyperosmolar medullary interstitium (up to 1200 mOsm/kg). The thin descending limb is permeable to water but impermeable to solutes, whereas the thick ascending limb actively pumps Na^+ , K^+ , $2Cl^-$ via NKCC2 cotransporter impermeable to water.

Clinical Pearl: Inulin clearance equals GFR (neither secreted nor reabsorbed); PAH clearance estimates Effective Renal Plasma Flow (ERPF) due to near-complete tubular secretion.

	Clearance Relation to GFR	Clinical Diagnosis
	Clearance = GFR	Gold standard measurement
	Clearance slightly overestimates GFR	Everyday clinical use
	Clearance = 0 mL/min (normally)	Glycosuria occurs